



FreshCart

E-Commerce Frontend Architecture

HTML5

CSS3

Vanilla JS

LocalStorage

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Project Overview



\$7T+

Global e-commerce market
projected by 2025

Technical Barrier

Full-stack dev requires servers, DBs, and backend knowledge many students lack.

Infrastructure Cost

Hosting and maintenance prevents experimentation in resource-limited settings.

Educational Gap

No unified tool shows e-commerce logic without backend complexity overhead.

Solution:

FreshCart — A complete e-commerce frontend built with HTML5, CSS3 & Vanilla JS, using LocalStorage for persistence. Zero backend required.

3-Layer System Architecture



01

Presentation Layer

HTML5 + CSS3

Semantic HTML5 markup · CSS Grid & Flexbox layouts · Responsive mobile-first design · 6 core pages



02

Application Logic Layer

Vanilla JavaScript ES6+

auth.js — Authentication · cart.js — Cart CRUD · products.js — Catalog · checkout.js & storage.js



03

Persistence Layer

Web Storage API

LocalStorage key-value store · JSON serialization · User-scoped keys (fc_cart_userId) · Cross-session data retention

Key Features & Modules



User Auth

Register, login, logout. bcrypt.js hashing with 10 salt rounds. Route guards.



Product Catalog

Browse, search & filter products. Category navigation. Dynamic DOM rendering.



Shopping Cart

Add/remove items, update quantities. Real-time total calculation. Per-user storage.



Checkout

Shipping form, payment method selection. Order placed & cart cleared.



Order History

View past orders per user. Status tracking. Persistent across sessions.



Responsive UI

Mobile-first CSS Grid. 4 → 2 → 1 col breakpoints. Clean modern UX.

Core Algorithms & Logic



User Registration

$O(n)$

- ✓ Validate inputs (len, match)
- ✓ Linear search for uniqueness
- ✓ bcrypt.hashSync (10 rounds)
- ✓ Store user → auto-login

Login Algorithm

$O(n)$

- ✓ Find user by username $O(n)$
- ✓ bcrypt.compareSync hash check
- ✓ Set currentUser in LocalStorage
- ✓ Redirect to home

Cart Management

$O(m)$

- ✓ Auth guard → get cart $O(1)$
- ✓ Search existing item $O(m)$
- ✓ Add new / increment qty
- ✓ Recalculate total $O(m)$

Checkout Flow

$O(m)$

- ✓ Validate cart not empty
- ✓ Validate shipping fields
- ✓ Create Order object + timestamp
- ✓ Save order → clear cart

Testing & Evaluation



24

Test Cases

Executed

100%

Pass Rate

All modules

<1s

Load Time

TTL measured

0.5ms

LocalStorage

Read/write avg

Module	Tests	Pass	Edge Cases Covered
Authentication	8	8	Duplicate username, wrong password
Cart Management	9	9	Empty cart checkout, qty updates
Checkout Flow	7	7	Missing fields, empty cart block

Browser Compatibility:



Chrome 120+



Firefox 120+



Safari 17+



Edge 120+

LocalStorage Data Model



USER

id (PK)

username (UQ)

passwordHash

email

fullName

phone

+ more...

PRODUCT

id (PK)

name

price

imageUrl

description

category

+ more...

CART

id (PK)

userId (FK)

items: CartItem[]

totalPrice

ORDER

id (PK)

userId (FK)

items: OrderItem[]

totalAmount

status

orderDate

+ more...

Storage Keys: fc_users · fc_currentUser · fc_cart_{userId} · fc_orders_{userId}

Limitations & Future Work

Limitations

! LocalStorage vulnerable to XSS attacks

! 5–10MB storage quota limits scale

! No multi-device account sync

! No real payment processing

! Client data editable via DevTools

! No service worker / true PWA

Future Enhancements

✓ Node.js/Express + MongoDB backend

✓ Stripe / PayPal payment integration

✓ IndexedDB for larger datasets

✓ Service Workers (offline PWA)

✓ Admin dashboard for management

✓ Dark mode & multi-language (i18n)



Conclusion

FreshCart proves that sophisticated e-commerce applications can be built with fundamental web technologies when properly architected.

- ✓ All 5 primary & 5 secondary objectives achieved
- ✓ 100% test pass rate across 24 test cases
- ✓ Modular architecture ready for backend integration
- ✓ Educational value for students learning e-commerce